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SPECIFICATION — Model Playground (PyQt5 + uv)

- **Status:** v0.1 — draft for implementation
- **Language:** Python 3.12 | GUI: PyQt5 | HTTP: httpx | Schema: jsonschema
- **Curriculum source:** `curriculum/week1/chapter1.md` (§12 Model Playground, §13 Measuring Tokens, §14 Comparing Outputs, §15 Enforcing Structured JSON).
- **Scope of this document:** the *authoritative specification* of an AI-native system. It is written to Level 2–3 (structured, mostly executable): behavior, interfaces, invariants, edge cases, and failure semantics are made explicit so an agent (or engineer) can derive implementation **and** verification with minimal inference.
- **Principle:** requirements express *intent*; this specification *operationalizes* intent into observable behavior plus the conditions under which we know it is correct.

---

0. Intent and purpose

Build a small desktop application that treats **different LLMs as interchangeable computational components** sitting behind one common interface, so that an engineer can run the same prompt through several models **side by side** and inspect, per model, not just the text but the *inference-substrate* properties that actually determine application quality:

- **latency** — decomposed into time-to-first-token (TTFT) and total latency;
- **throughput** — generation rate (tokens/second, TPS);
- **economics** — token usage and derived *cost*. Locally there is no vendor bill, so cost is a configurable per-1k-token field for bookkeeping that defaults to `0`; the machinery is identical to the hosted case (chapter §13).
- **reliability** — does the output parse, and does it satisfy a schema?

This directly instantiates the chapter's core thesis: *`AI Application = Probabilistic Components + Deterministic Systems`*. The model supplies the probabilistic computation (§17); everything here is the **deterministic boundary** — the "harness layer" (§1) and the "reliability boundary" (§15) — that the model is never trusted to provide on its own.

Deployment decision (chapter §3, APIs vs. Local Models): inference is *local*, performed by the **Ollama** runtime (default `http://localhost:11434`) against models already pulled on the machine. The app talks to Ollama's HTTP API (a small `localhost` HTTP call, not a hosted cloud API) and the model runs on the host's CPU/GPU/NPU. This is the "local inference" architecture of chapter §3: the app gains control over latency, privacy, availability, version, and configuration, and pays the corresponding engineering burden (owning runtime availability, model download, memory/thermal constraints — see E-13/E-14/E-15).

Primary product surface: not "chat with a model," but a **side-by-side evaluation panel** in which, for one prompt, the user sees $M_1(x), M_2(x), \ldots$ with each model's metrics and — optionally — its validated structured output.

Non-goals (explicit, to constrain the solution space):

- No conversation history / multi-turn memory beyond the current prompt (§6 says history is a *context* input, not a feature to build here).
- No tool calling / function-call execution loop (§7 is conceptual; out of scope for this minimal substrate). *See open question Q-04.*
- No multimodal inputs (§8 is conceptual; text-only here).
- No persistence of runs, no cloud/hosted provider, no auth. Ollama is a localhost daemon with no key (E-13). Model **download/pull** is out of scope — the app lists and runs *already-pulled* local models (§3 local-inference model, E-14).
- No alternative "quality" LLM-as-judge scoring: comparison here is *observational* (metrics + human/JSON inspection), not an automated judge.
- **The app must not *require* Ollama to import, build, or run its test suite:** a deterministic `MockModel` **test double** provides every capability for offline/CI use (§9). Ollama is the *real* backend; `MockModel` is what the test suite drives.

---

1. Actors and goals

Actor	Goals
User (human, single process)	Enter one prompt; pick a set of models + generation parameters; run them in parallel or sequentially; compare outputs, metrics, and costs side by side; optionally obtain and inspect a validated structured output.

Actor	Goals
Model (`Model` implementation: `OllamaModel` for the real run, `MockModel` as a test double)	Given messages + params, produce a (streaming) text completion with usage stats, or fail — **never** touch the GUI.
Ollama daemon *(external)*	Local inference runtime on `localhost:11434` (`/api/chat`, `/api/tags`). Owns the model weights and the CPU/GPU/NPU. Not part of this project; E-13/E-14 handle its absence.
Registry (`ModelRegistry`)	Map a model identifier to a configured model + its pricing, so the rest of the system never branches on provider. For Ollama, populated from `GET /api/tags`.
Metrics layer (`metrics.py`)	Turn a run's raw timings + usage into TTFT / latency / TPS / cost. Pure, headless, testable.
Structured layer (`structured.py`)	Turn a raw text completion into a *validated* typed object via parse → validate → accept/reject, with a retry/fallback policy. Pure, headless, testable.
Worker (`RunWorker(QThread)`)	Drive one model's generation off the UI thread; emit stream chunks and a final result; crash cleanly.
UI (`MainWindow`)	Present controls + a grid of per-model result panels; never block on inference.

---

2. Requirements (intent, high level)

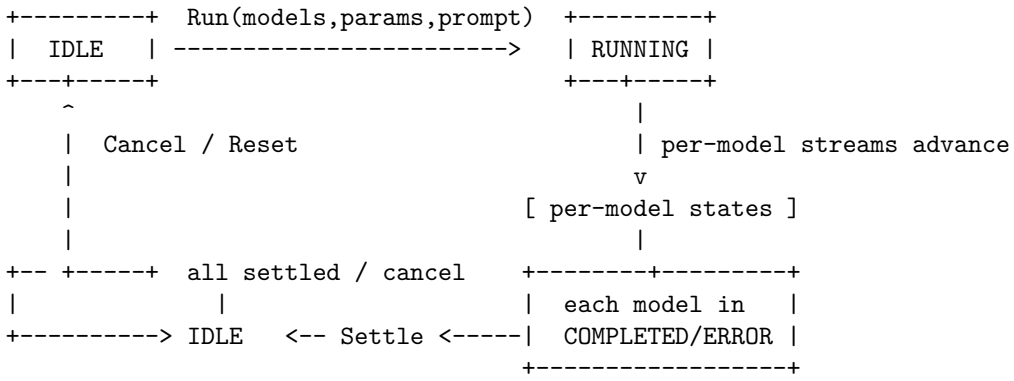
ID	Statement
R-01	The application shall expose a **common model interface** (`Model`) such that a model can be substituted for another without changing any other layer of the system.
R-02	The user shall specify, per run: a **list of one or more models** (chosen from the locally-pulled Ollama models, §R-16, or from the built-in `MockModel` registry when Ollama is unavailable), a **generation-parameter set** (temperature, top_p, max_tokens), and the **prompt** (as a message list).
R-03	For each selected model the app shall display the **generated text** (fully, or streaming token-by-token).
R-04	For each model run the app shall display **metrics**: `tput latency (ms)`, `ttft_ms`, `tps`, and `status`.
R-05	For each model run the app shall record **token usage**: `prompt_tokens`, `completion_tokens`, `total_tokens`. For Ollama these come from the response envelope (`prompt_eval_count`, `eval_count`); for `MockModel` from its deterministic tokenizer.
R-06	For each model run the app shall display a **cost** computed from usage and that model's registered per-1k-token prices. For local Ollama prices default to `0` (no vendor bill); the field is configurable so the same machinery models a nominal compute cost (chapter §13).
R-07	The app shall run the selected models **side by side**, i.e. the user can compare $M_1(x), ..., M_k(x)$ from one prompt in a single view (§14).
R-08	Streaming mode shall show the response as it arrives and must report $TTFT \leq T_{complete}$ and distinguish them (§12.2, §11).
R-09	When **structured output** is requested, the app shall emit a **validated typed object** (not raw prose): `{ answer, confidence, reasoning_required }` (or the configured schema).
R-10	Structured output shall enforce the pipeline **raw → parse → validate → accept/reject**; never assume "valid-looking ≥ *valid*" (§6, §15).

ID	Statement
R-11	On a parse/validation failure the app shall apply a **retry-then-fallback** policy and surface the failure state, never silently fabricate a valid object (§15).
R-12	The GUI must remain responsive while one or more models generate (§3 streaming).
R-13	All model I/O must occur off the Qt main (event-loop) thread.
R-14	The project shall be reproducible via `uv` on Python 3.12, and runnable/testable **fully offline (no Ollama)** via `MockModel`; the real GUI uses Ollama when reachable and degrades gracefully when it is not (E-13).
R-15	A fixed `seed` + `temperature=0` shall make a run **deterministic**: `MockModel` is *bitwise* reproducible; `OllamaModel` is reproducible *best-effort* (Ollama honors `options.seed`, but floating-point/kernel differences across runs may perturb later tokens — token *counts and metrics* are asserted, bitwise text is not).
R-16	On launch the app shall **discover locally-pulled Ollama models** via `GET /api/tags` and populate the model checklist from that list; if Ollama is unavailable it shall fall back to the built-in `MockModel` registry and state so in the UI (E-13).
R-17	A model that is not pulled locally shall not be runnable: selecting it yields a per-panel `ERROR` (`model not found / pull required`) rather than a silent failure or a hard crash (E-14).

---

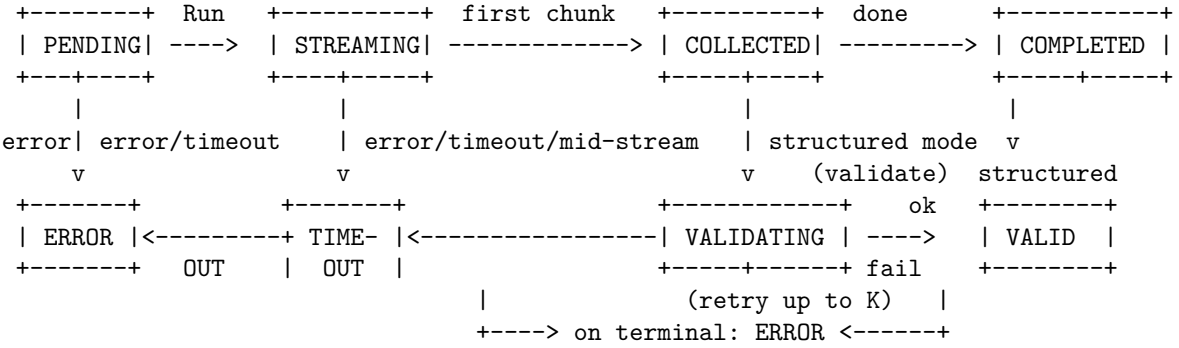
3. Behavior and state model

3.1 System state machine (`RunState`, per comparison run)



Overall run state is an **aggregate** of the per-model states. A run is settled when *every* model has reached `COMPLETED` or a terminal error. A single model failure never aborts the others.

3.2 Per-model run state machine (`ModelRunState`)



State	Meaning	Terminal?
`PENDING`	Scheduled; request not yet issued.	no
`STREAMING`	Chunks arriving; `text` is a partial accumulation.	no
`COLLECTED`	Full text + usage available (streaming done, or non-streaming returned).	no
`VALIDATING`	Structured mode: parse + schema-validate the collected text; may retry.	no
`COMPLETED`	Plain-text mode terminal success (no structured request).	yes
`VALID`	Structured mode: a validated typed object exists.	yes
`ERROR`	Terminal fault: request error, timeout, mid-stream failure, or all retries exhausted (§8).	yes
`TIMED_OUT`	A dedicated terminal when the total elapsed exceeds `timeout_s`.	yes
`CANCELLED`	User cancelled the run while this model was active.	yes

Transition rules: - Exactly **one** attempt sequence per model per run; within it, structured mode may perform up to `max_retries` parse attempts (§8). Each retry re-issues generation with an *error-informed* prompt (original prompt + a directive describing the last parse/validation failure). - A model reaching `ERROR`/`TIMED_OUT`/`CANCELLED` is terminal for *that* panel but leaves all sibling panels running. - The overall run reaches `IDLE` (settled) only when **every** panel is terminal. - `Stop`/`Cancel` from `IDLE` while a run is active tears every model's worker down (no live worker survives; see I-010).

3.3 Threading model

- **UI thread (main):** widget event loop + all panel updates only.
- **Worker thread(s):** one `RunWorker(QThread)` **per model**, each owning its `Model` and its `generate`/`stream` call. Never touch widgets.
- Communication is Qt signals only (`QThread` + `pyqtSignal`), plus a per-worker thread-safe cancel flag guarded by a lock. No shared mutable buffers cross threads except via queued (auto-connect) signals.
- **Concurrency policy:** by default all selected models stream **concurrently** (one worker each). A `sequential` mode (one-at-a-time) is allowed via UI flag to keep concurrency = 1; default is **concurrent**.

3.4 Streaming vs. non-streaming

The common interface offers **both**; the UI picks one:

- **Streaming:** the model yields `StreamChunks`; `ttft` is measured to the **first non-done chunk** that carries a non-empty delta. Ollama emits **NDJSON** — one JSON object per line `{"message": {"role","content"}, "done": bool, …}`; `OllamaModel` splits the response on newlines and parses each line; the **final** line has `done: true` and carries `prompt_eval_count` / `eval_count` (the C-01 `Usage`). `MockModel` yields equivalent `StreamChunks` with no HTTP.
- **Non-streaming:** `OllamaModel` sets `stream=false` and parses the single JSON object; the interface returns one `ModelResponse`; `ttft = total_latency` (a special-case defined explicitly so the metric is always total-ordered, E-05).
- **Determinism:** generation passes `options.seed` and `temperature`; with `temperature=0` + fixed `seed` Ollama output is reproducible (R-15, best-effort).

4. Interfaces / contracts

C-01 Core types

```
class Role: SYSTEM = "system"; USER = "user"; ASSISTANT = "assistant"

@dataclass
class Message:
    role: str        # one of Role
    content: str

@dataclass
class GenerationParams:          # part of "generation parameters" the model needs
    temperature: float = 0.0     # in [0, 2]; 0 => deterministic-ish
    top_p: float       = 1.0     # in (0, 1]
    max_tokens: int    = 512     # > 0
    seed: int | None   = None    # fixed => reproducible when the model supports it (R-15)

@dataclass
class Usage:
    prompt_tokens: int     # >= 0
    completion_tokens: int# >= 0
    @property def total_tokens(self) -> int ...    # = prompt + completion (I-001)

@dataclass
class StreamChunk:
    delta: str             # text delta; may be "" on a keepalive
    finished: bool         # True on the final chunk
    usage: Usage | None    # set (or updated) on the final chunk; None on mid-stream

@dataclass
class ModelResponse:          # the non-streaming return, and the *collected* form
    text: str
    usage: Usage
    model_id: str
```

C-02 `Model` — the minimal interchangeable interface (R-01)

```
class Model(ABC):
    @property
    def model_id(self) -> str: ...              # stable id, e.g. "/ollama/llama3.2", "mock/fast"

    @abstractmethod
    def generate(self, messages: list[Message], **params) -> ModelResponse:
        """Non-streaming. Precondition: messages non-empty.
        Postcondition: return a ModelResponse whose usage.prompt_tokens >= 0.
        May raise on transport/timeout errors (caught by the worker -> ERROR)."""

    @abstractmethod
    def stream(self, messages: list[Message], **params) -> Iterator[StreamChunk]:
        """Streaming. Yields >=1 chunk; the LAST chunk has finished=True and
        carries the final usage (I-008)."""
```

`**params` accepts `GenerationParams` fields. **Invariance (I-002):** no layer other than a concrete `Model` implementation (`OllamaModel`, `MockModel`) may hard-code a provider endpoint, model name, or provider-specific request shape. All other modules see only `Model` + `Message` + `GenerationParams`.

C-03 `ModelRegistry`

```python
@dataclass
class ModelSpec:
    model: Model
    label: str | None = None              # display name; defaults to model.model_id
    price_input_usd_per_1k: float = 0.0
    price_output_usd_per_1k: float = 0.0

class ModelRegistry:
    def __init__(self) -> None ...
    def register(self, spec: ModelSpec) -> None
    def get(self, model_id: str) -> ModelSpec      # KeyError if unknown
    def available(self) -> list[ModelSpec]
    def cost_usd(self, model_id: str, usage: Usage) -> float
```

Invariance (I-003): pricing is bound to the model **in the registry**, not hard- coded in the UI or the worker. For **local Ollama**, `price_output_usd_per_1k` and `price_input_usd_per_1k` default to `0.0` (no vendor bill, R-06); the fields remain so a nominal compute cost can be booked in.

C-03b `OllamaClient` (thin transport for `OllamaModel`, `model.py`)

A tiny `httpx`-based client over the Ollama HTTP API. It is the *only* place that knows URLs/JSON shapes (I-002). No import of a vendor SDK; `httpx` is a generic HTTP client.

```python
OLLAMA_BASE = "http://localhost:11434"   # overridable via env OLLAMA_HOST

class OllamaClient:
    def __init__(self, base: str = OLLAMA_BASE, timeout_s: float = 30.0) -> None ...

    def list_models(self) -> list[str]:
        """GET /api/tags -> [name, ...] of *locally-pulled* models (R-16, E-13).
        Raises ConnectionError-like on unreachable daemon; the caller treats that as
        "Ollama unavailable" and falls back to MockModel (E-13)."""

    def chat(self, model: str, messages: list[Message],
             params: GenerationParams, stream: bool) -> "Iterator[StreamChunk] | ModelResponse":
        """POST /api/chat. Maps GenerationParams -> Ollama `options`
        (temperature, top_p, num_predict=max_tokens, seed). Stream: NDJSON, last
        line carries prompt_eval_count/eval_count -> Usage (R-05, I-008).
        Raises on HTTP/network error or malformed NDJSON (E-02/E-07/E-15)."""
```

Mapping (GenerationParams -> Ollama `options`): `temperature`→`temperature`, `top_p`→`top_p`, `max_tokens`→`num_predict`, `seed`→`seed` (omit when `None`). **Endpoint for unknown model:** Ollama returns HTTP 404/error for a model not pulled locally; `OllamaModel` surfaces that as `KeyError`/panel `ERROR` (R-17, E-14).

C-04 Metrics (`metrics.py`, pure — testable headless)

```python
@dataclass
class RunMetrics:
    model_id: str
```

```
    status: str
    ttft_ms: float                # ms to first non-empty delta (or total latency if non-streaming)
    total_latency_ms: float
    tps: float                    # completion_tokens / generation_seconds
    usage: Usage
    cost_usd: float
    retries: int = 0              # structured-mode parse retries that occurred
    error: str | None = None

def compute_metrics(
    model_id: str,
    *,
    t_request: float, t_first_token: float | None, t_complete: float,
    usage: Usage, price_input: float, price_output: float,
    retries: int = 0, status: str = "COMPLETED", error: str | None = None,
) -> RunMetrics
```

Definitions (enforced by I-004, I-005, I-006):

$$T_{\text{ttft}} = t_{\text{first token}} - t_{\text{request}} \qquad T_{\text{complete}} = t_{\text{complete}} - t_{\text{request}}$$

$$\text{TPS} = \begin{cases} \dfrac{\text{completion\_tokens}}{T_{\text{complete}} - T_{\text{ttft}}} & \text{if } T_{\text{complete}} > T_{\text{ttft}} \\ 0.0 & \text{otherwise (guard, I-005)} \end{cases}$$

$$\text{cost\_usd} = \frac{\text{prompt\_tokens}}{1000} P_{\text{in}} + \frac{\text{completion\_tokens}}{1000} P_{\text{out}} \qquad \text{(C-13 unit economics)}$$

- **Cost per successful task** (§13) is a *derived* aggregate the UI shows: total cost across a run divided by the count of models that reached a terminal success state (`COMPLETED` or `VALID`). If that count is 0, the denominator is 1 and the value is the raw total (never division by zero; I-007).

C-05 Structured output (`structured.py`, pure — testable headless)

Target schema (the default "answer" schema of §15):

```
{
  "type": "object",
  "additionalProperties": false,
  "required": ["answer", "confidence", "reasoning_required"],
  "properties": {
    "answer":              {"type": "string", "minLength": 1},
    "confidence":          {"type": "number", "minimum": 0, "maximum": 1},
    "reasoning_required":  {"type": "boolean"}
  }
}
```

```
@dataclass
class ValidationResult:
    ok: bool
    data: dict | None       # populated iff ok
    errors: list[str]       # human-readable reasons; empty iff ok
    raw: str                # the exact text that was parsed

def parse_json(text: str) -> tuple[dict | None, list[str]]:
```

```
    """Strip a single ```json …``` fence if present, then json.loads.
    Returns (None, [reason]) on any failure - never raises."""

def validate(data, schema) -> ValidationResult:
    """jsonschema.validate; collect every error message. Never raises."""
```

Pipeline (`structured.py`):

```
raw text --> strip optional code fence --> json.loads --> jsonschema.validate --> dict
         \_____________ any step fails => attempt retry (up to max_retries),
                        else terminal ERROR with the first failure reason (E-03)
```

C-06 `RunWorker(QThread)`

```
signals:
    token(model_id: str, delta: str)      # per streamed chunk (UI appends)
    metrics_ready(model_id: str, RunMetrics)  # emitted once, on settle or terminal
    crashed(model_id: str, message: str)  # uncaught worker fault (E-07)

public methods (thread-safe to call from main):
    start_run(...) -> None               # kick off the model's generate/stream
    cancel()     -> None                 # idempotent; safe from any state
```

Each `RunWorker` is 1:1 with one model panel. A comparison run owns a `QThreadList`-style container managing N workers concurrently.

C-07 Data structures carried to the UI

```
ModelPanelView:
   model_id, label, status, text (accumulated),
   metrics: RunMetrics | None,
   structured: ValidationResult | None,
   streaming: bool, done: bool
```

---

5. UI specification

5.1 Top-level layout (`MainWindow`)

```
+-------------------------------------------------------------------------+
| Title: Model Playground - an inference substrate                        |
+--------------------------+----------------------------------------------+
| LEFT COLUMN (controls)   | RIGHT COLUMN: grid of per-model panels       |
|  Prompt (QPlainTextEdit) |   +-------------------------------+          |
|  system prompt (opt.)    |   | [Mock Fast]   RUNNING         |          |
|  Temperature / top_p /   |   |  text: "..." (streaming)      |          |
|  max_tokens / seed       |   |  TTFT __ ms  Lat __ ms  TPS _|          |
|  Models: checklist of    |   |  in __/out __   cost $0.000   |          |
|   registered models (>=1)|   |  [structured: OK/FAIL badge] |          |
|  [stream] checkbox       |   +-------------------------------+          |
|  [structured] checkbox   |   +-------------------------------+          |
|  [sequential] checkbox   |   | [Mock Slow]   COMPLETED       |          |
|  [ Run ][ Cancel ]       |   |  text: "..."                  |          |
|  state label             |   |  TTFT __  Lat __  TPS _ $__   |          |
|  cost-per-task summary   |   +-------------------------------+          |
+--------------------------+----------------------------------------------+
```

C-08 Control widgets and validation

Widget	Type	Constraint	Invalid behavior (§8)
Prompt	`QPlainTextEdit`	non-empty	disable Run; message `Prompt must not be empty`
Temperature	`QDoubleSpinBox`	$[0, 2]$	clamp
top_p	`QDoubleSpinBox`	$(0, 1]$	clamp; message if out of range
max_tokens	`QSpinBox`	$1 \ldots 100{,}000$	message if < 1
seed	`QLineEdit(int, optional)`	integer or blank	non-integer -> message + disable Run; blank => `None`
Models	`QCheckBox` per registered model	≥ 1 selected	disable Run; message `Select at least one model`
Run / Cancel	`QPushButton`	enabled per state table (§3.1)	–
stream / structured / sequential	`QCheckBox`	–	–

5.2 Panels

- Each panel shows: label, status pill, growing text, the four C-04 metrics, usage in/out, cost, and — when structured mode is on — a pass/fail badge plus the parsed JSON or the collected validation errors.
- Streaming: text appends as `token` signals arrive; metrics update once on settle.

---

6. Invariants (must hold in every valid implementation)

ID	Invariant	Verified by
I-001	`Usage.total_tokens == prompt_tokens + completion_tokens`; both ≥ 0.	T-01
I-002	No module outside a concrete `Model` implementation imports a provider-specific SDK or names a concrete model. (Architecture; checked by T-02 import scan.)	T-02
I-003	Pricing lives only in `ModelRegistry`; `cost_usd` is derived, never hard-coded in UI/worker.	T-03
I-004	`ttft_ms <= total_latency_ms` for **every** finished run (incl. the non-streaming special case `ttft = total`).	T-04
I-005	`tps >= 0`; the zero-generation-duration case yields `tps == 0.0` (never `inf`/`nan`).	T-04
I-006	`cost_usd` equals the C-04 formula exactly (within float epsilon) for the supplied prices.	T-05
I-007	`cost_per_success_task` is never a division by zero; all-failed run uses denominator 1.	T-06
I-008	Every `stream` yields a final chunk with `finished=True`; that chunk's `usage.completion_tokens` equals the streamed token count and equals the non-streaming `generate` usage for the same input+seed (mock, R-15).	T-07

ID	Invariant	Verified by
I-009	No valid-looking output is accepted: a panel reaches `VALID` **only** via `validate(...).ok == True`. A syntactically valid but out-of-range/missing-field object must NOT validate.	T-08
I-010	At most one live worker per model panel; `cancel()` leaves zero live workers.	T-09
I-011	All heavy inference is off the UI thread (worker thread id != main thread id).	T-11
I-012	Fixed `seed` on the mock model yields a bitwise-identical `text`, `usage`, and metrics for the same prompt + params.	T-07

---

7. Constraints (precise and measurable)

ID	Constraint	Measurement
K-01	While `RUNNING` with all models streaming, a posted task on the event loop is serviced within $p_{95} < 50$ ms.	T-11 (offscreen)
K-02	A mid-stream failure or timeout leaves the panel in a well-defined terminal `ERROR`/`TIMED_OUT`; partial text is preserved for display.	E-02, T-10
K-03	Default `max_retries = 2`; default per-run `timeout_s = 30`.	T-08
K-04	No provider SDK or key is required to import, run, or test the app offline.	T-02, T-14

---

8. Edge cases and failure semantics

ID	Situation	Required behavior
E-01	No model selected / empty prompt	Run disabled; inline message; no worker spun.
E-02	A model fails **mid-stream** (raises inside the iterator, or times out)	Panel -> `ERROR`/`TIMED_OUT`; partial text kept; **siblings continue**; the run still settles. Never fabricate completion.
E-03	Structured parse or validation fails	Retry up to `max_retries` with an error-informed prompt; on exhaustion -> `ERROR` showing the first failure reason. **Never** present an unvalidated dict as `VALID` (I-009).
E-04	Streaming disabled (non-streaming path)	`ttft_ms == total_latency_ms`; `tps` computed over the single interval (guards as in I-005).
E-05	Zero completion tokens (empty response)	`tps == 0.0` (I-005); not `inf`/`nan`; status may still be `COMPLETED`.
E-06	`max_tokens = 0` or other invalid param	Validation (§5.2) blocks Run; message shown.
E-07	Uncaught worker exception	Emit `crashed(model_id, msg)`; panel -> `ERROR`; **never** abort the whole process.

ID	Situation	Required behavior
E-08	`Cancel` while active	Every worker torn down and joined; no live worker survives (I-010); clean exit.
E-09	Unknown model id requested	`ModelRegistry.get` raises `KeyError` -> surfaced as panel `ERROR`, not a crash.
E-10	No display (CI)	Tests use `QT_QPA_PLATFORM=offscreen`; logic unaffected.
E-11	Code-fence-wrapped JSON	`parse_json` strips a single json … fence before `loads`; bare JSON also accepted.
E-12	Duplicate concurrent runs from the UI	Cancel any prior live run first; exactly one active run at a time (I-010).
E-13	Ollama daemon unreachable (not running / `OLLAMA_HOST` wrong)	`OllamaClient.list_models()` raises; on launch the UI **falls back to the `MockModel` registry** with a status banner `Ollama unavailable - using mock models` (R-16); no crash, no hang.
E-14	Model not pulled locally (Ollama returns HTTP 404/`model not found`)	That panel -> `ERROR` with `model not found: 'foo' - pull it with ollama pull`. Siblings continue; no crash (R-17/E-09).
E-15	Malformed NDJSON line / missing `eval_count` in an Ollama stream	Skip non-JSON keepalive lines; on a malformed final line, keep partial `text` and set `usage` to best-effort `count_tokens` on the accumulated text (or `0`) rather than crash; surface a `warning` on the panel (E-02).

Failure philosophy (spec-engineering doctrine): the model is the *probabilistic* side of the reliability boundary (§15); everything in §4 C-04/C-05 is the *deterministic* side. The dominant failure here is **accepting an unvalidated artifact as valid** and **letting one model's failure poison the others**. This spec forbids both: I-009 enforces the validate gate; E-02/E-07 isolate faults per panel.

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9. Acceptance criteria, tests, and evals

All tests target Level-3 executable criteria. Pure layers (C-04, C-05) and mock models need **no Qt and no network**; GUI tests run offscreen via `pytest-qt`.

9.1 Model interface + mock model (deterministic, no Qt/network)

ID	Criterion
T-01	**Usage arithmetic (I-001):** `Usage(p,c).total_tokens == p + c` for representative pairs.
T-02	**Interface purity (I-002 / K-04):** a source scan asserts the *only* provider-aware module is `OllamaClient`; no module imports a vendor SDK. `httpx` is allowed only inside `OllamaClient`. The app imports + runs with **no Ollama and no keys** (via `MockModel`).
T-07	**Reproducibility (I-012 / I-008 / R-15):** `MockModel` with `seed=42` yields bitwise-identical `text` and `usage` across two runs; its streamed token count equals its non-streaming `usage.completion_tokens`. Determinism of TTFT/latency is not asserted (wall-clock), but token counts are.

9.2 Metrics (pure) — fast, run always

ID	Criterion
T-03	**Pricing location (I-003):** changing a model's registry price changes `cost_usd` in the UI/worker with no other edit.
T-04	**TTFT ordering & TPS guard (I-004, I-005):** `ttft_ms <= total_latency_ms`; zero-generation-duration => `tps == 0.0` (not `inf`/`nan`).
T-05	**Cost formula (I-006):** `cost_usd` matches $N_{in}P_{in} + N_{out}P_{out}$ exactly for non-trivial prices.
T-06	**Cost-per-task (I-007):** all-failed run divides by 1; mixed run divides by the success count.

9.3 Structured output (pure) — the reliability boundary

ID	Criterion
T-08	**Validate gate (I-009 / R-10):** (a) a conforming object validates `ok`; (b) an out-of-range `confidence` (e.g. `1.5`) **fails**; (c) a missing required field **fails**; (d) extra property **fails** (`additionalProperties:false`); (e) after `max_retries` exhaustion the panel is `ERROR`, never `VALID`.
T-14	**Fence + bare JSON (E-11):** `parse_json` accepts both `'''\n{...}\n'''` and bare `{...}`; both route to the same schema; a non-JSON string returns `(None, reason)`.

9.4 GUI / integration (offscreen, pytest-qt)

ID	Criterion
T-09	**Single worker / cancel (I-010 / E-12):** starting a run while one is active cancels the prior; after `cancel()` zero workers are alive.
T-10	**Isolated failure (E-02 / K-02):** with two mock models, one forced to raise mid-stream, that panel is `ERROR` while the sibling is `COMPLETED`; the run settles.
T-11	**Off-thread + liveness (I-011 / K-01):** while a run streams offscreen, a posted task is serviced < 50 ms; the worker thread id != the main thread id.
T-13	**State machine (§3.1/§3.2):** idle -> run -> (all settle) -> idle; cancel from running reaches idle with terminal panels; control-enable flags match §3.1.
T-15	**Validation UI (E-01):** empty prompt and zero models selected disable Run with the inline messages.

9.5 Manual / smoke eval (not automated; recorded)

- Launch with `uv run model-playground`; confirm 2–3 models side by side: one **local Ollama** model (e.g. `mock/fast`, or a real `/api/tags` model if Ollama is running), one slow, one misbehaving (raises), and observe independent panels, streaming text, metric columns, and a structured pass/fail badge (E-02, qualitative).
- Confirm the Ollama-unreachable banner: with no daemon, the checklist falls back to the `MockModel` registry (E-13/R-16).

9.6 Ollama client (integration, no Qt; network-stubbed)

ID	Criterion
T-16	**Discovery + fallback (R-16 / E-13):** with `OllamaClient` pointed at an unreachable host, `list_models()` raises a connection error and the app falls back to `MockModel` — asserted by a `conftest` fixture that forces `OLLAMA_HOST` to a dead port; **no network** is real.
T-17	**NDJSON + param mapping (C-03b / I-008 / E-15):** using `httpx.MockTransport` (no real network), `OllamaClient.chat` (a) maps `max_tokens`→`num_predict`, `seed`→`seed`, `temperature`/`top_p` through; (b) parses a multi-line NDJSON stream into `StreamChunk`s whose final chunk carries `finished=True` and `usage` from `eval_count`/`prompt_eval_count`; (c) keeps partial text and best-effort usage when a trailing line is malformed.
T-18	**Unknown model (R-17 / E-14):** a mock `404`/`model not found` response yields panel `ERROR` with the `model not found: '…'` message; it does not crash.

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10. Dependencies and environment

Concern	Decision	Rationale
Package/env manager	**uv**	Fast, reproducible, pins Python; satisfies R-14.
Python	**3.12** ($\geq 3.12, < 3.13$)	Requested; stable PyQt5.
GUI	**PyQt5 (5.15)**	Requested
HTTP	**httpx** (stream-capable)	Thin client for `OllamaModel` (C-03b) over the Ollama localhost API; **not** imported by pure layers (metrics/structured) nor by `MockModel`; not required to run tests (K-04).
Local inference engine	**Ollama** *(external)*	Runtime at `http://localhost:11434` (§0 deployment decision). Not a Python dependency — a host prerequisite. The app degrades to `MockModel` when it is absent (E-13).
Schema validation	**jsonschema**	C-05 `validate`.
Mock model	**in-repo** `MockModel`	Deterministic, offline, reproducible (R-15); the default test/runtime double; provides streaming + a "slow" and a "raising" variant for E-02/E-07.
Dev deps	**pytest, pytest-qt, ruff**	Automated Level-3 tests (§9) + lint.
GUI test backend	`QT_QPA_PLATFORM=offscreen`	Headless CI (E-10).

Reproducibility (see `README.md`):

```
# host prerequisite for the real path (optional; the mock path needs neither):
#   ollama pull llama3.2
uv sync                          # create .venv (Python 3.12), install everything
uv run pytest                    # run the §9 suite (fully offline; no Ollama needed)
uv run model-playground          # launch GUI (Ollama if reachable, else mock models)
```

11. Traceability matrix (id -> where realized)

- R-01 –> C-02(Model), C-03(Registry) –> T-02
- R-02 –> C-01(GenerationParams), C-08 controls –> T-15
- R-03 –> C-06(token signal), C-07 panel –> T-11
- R-04 –> C-04(RunMetrics) –> T-04
- R-05 –> C-01(Usage), I-001 –> T-01
- R-06 –> C-03(cost_usd), I-003/006 –> T-03, T-05
- R-07 –> §5.1 grid, C-06 N workers –> T-10, T-13
- R-08 –> C-04(ttft/tps), §3.4 –> T-04
- R-09/10 –> C-05(parse,validate), I-009 –> T-08
- R-11 –> §8 E-03, C-05 retry –> T-08
- R-14 –> §10, pyproject, K-04 –> T-02
- R-15 –> C-01(seed), MockModel –> T-07
- I-007 / §13 –> C-04 cost-per-task –> T-06
- I-010 / E-12 –> C-06(cancel), §3.2 –> T-09
- I-011 / K-01 –> §3.3 worker threads –> T-11
- E-02 / K-02 –> §3.2 per-model terminal –> T-10
- R-05 / R-15 –> C-01(Usage: Ollama eval_count / prompt_eval_count), I-001/008/012 –> T-01, T-07
- R-16 / E-13 –> C-03b(list_models), C-03 discovery, §8 –> (smoke, E-13)
- R-17 / E-14 –> C-03b(chat 404 -> ERROR) –> T-10-style
- C-03b / I-002 –> OllamaClient (only provider-aware module) –> T-02
- C-03 / I-003 –> local default price 0.0 –> T-05
- E-15 –> C-03b(NDJSON parse: keep partial + best-effort usage) –> (smoke)

Open questions / ambiguities flagged for the human (spec elicitation):

1. **Structured schema.** The default schema is the §15 "answer" object. Should it be *user-editable* in the UI, or fixed to this one object for v0.1? (spec: fixed for v0.1; Q-01).
2. **Pricing source.** Prices are hand-entered per model in the registry for v0.1. Should we pull a canonical price table? (Q-02 — recommend: hand-entered, documented as illustrative.)
3. **Default local model.** Which Ollama model(s) should the GUI pre-select for the first smoke run? (Q-03 - recommend: pre-select the first model returned by `/api/tags`, plus the built-in `mock/fast` and `mock/slow`.)
4. **Tool calling.** §7 (tool calling) is deliberately **out of scope** here, as is §8 (multimodal). Confirm these remain conceptual-only in v0.1. (Q-04)
5. **Concurrency default.** Concurrent (one worker/model) vs. sequential default. Spec picks **concurrent**, with a sequential checkbox. (Q-05 — confirm.)
6. **Retry semantics for non-structured mode.** Retries apply only to the structured pipeline; plain-text mode has no retry. Confirm. (Q-06)

End of specification. This document is the source of truth; implementation and tests are to be derived from it and kept in sync per §11.